Dinosaur Poster Key

U02 / S02.3

MENU FACTS · ALL 15

From the 2026-10-27 research report. Hip: L = lizard-hipped (saurischian), B = bird-hipped (ornithischian). Lengths in meters. Fossil sites list solid finds only.

DINOSAUR	PERIOD	MYA	HIP	DIET	LEGS	LENGTH	FOSSIL LOCATIONS
TRIASSIC · 252 TO 201 MYA · ALL LIZARD-HIPPED							
Herrerasaurus	Late Triassic	~231	L	Meat	2	3–6	Argentina
Eoraptor	Late Triassic	~231–228	L	Small animals, maybe plants (sources conflict)	2	~1	Argentina
Plateosaurus	Late Triassic	~214–204	L	Plants	2	~5–10	Germany, Switzerland, France
Liliensternus	Late Triassic	poorly dated; 228–202 across sources	L	Meat	2	~5	Germany
Coelophysis	Late Triassic (latest)	~206–201	L	Meat	2	2–3	USA: New Mexico
JURASSIC · 201 TO 145 MYA							
Dilophosaurus	Early Jurassic	~196–183	L	Meat	2	~6–7	USA: Arizona
Allosaurus	Late Jurassic	~155–147	L	Meat	2	~8.5 (largest ~9.7)	USA: Colorado, Utah, Wyoming; Portugal
Stegosaurus	Late Jurassic	~155–150	B	Plants	4	~7–9	USA: Colorado, Utah, Wyoming; Portugal
Diplodocus	Late Jurassic	~155–147	L	Plants	4	~24–26	USA: Colorado, Utah, Wyoming, Montana
Brachiosaurus	Late Jurassic	~154–153	L	Plants	4	~18–22	USA: Colorado, Utah, Wyoming, Oklahoma. No Africa dot.
CRETACEOUS · 145 TO 66 MYA							
Spinosaurus	Cenomanian (early Late Cretaceous)	~100–94	L	Mostly fish	2	~12–15	Egypt, Morocco; Niger (S. mirabilis, genus level); Algeria optional, marked “?”
Velociraptor	Late Cretaceous (Campanian)	~75–71 (tentative)	L	Meat	2	~1.5–2	Mongolia; China (Inner Mongolia)
Tyrannosaurus rex	Late Cretaceous (Maastrichtian)	~68–66	L	Meat	2	~12–13	USA: Montana, South Dakota, Wyoming; Canada: Saskatchewan
Triceratops	Late Cretaceous (Maastrichtian)	~68–66	B	Plants	4	~7–9	USA: Montana, North Dakota, South Dakota, Wyoming; Canada: Alberta, Saskatchewan
Ankylosaurus	Late Cretaceous (Maastrichtian)	~68–66	B	Plants	4	~6–8	USA: Montana, Wyoming; Canada: Alberta

GRADING NOTES

- **Period dates.** The menu uses rounded dates: 252–201, 201–145, 145–66 mya. The ICS values are 251.902, 201.4, 145.0 and 66.0 Ma. The researcher could not re-read the ICS chart this session, so those are **PROVISIONAL** until checked. Rounded dates are full credit either way.
- **Triassic hip types.** The whole Triassic row is lizard-hipped on purpose. No securely bird-hipped Triassic dinosaur is known. The best candidate, Eocursor, may really be Early Jurassic. Worth saying out loud: early dinosaurs were mostly lizard-hipped.
- **Hip type from a picture.** Velociraptor and birds are lizard-hipped, but their pubis points backward. A student who reads hip type off a Velociraptor skeleton will get it wrong. The student page says “in most of them” for this reason.

- **Habitat and special feature** are not on the menu and were not researched for this key. Accept any reasonable answer from the habitat list that the student can point to a source for. **PROVISIONAL**: there is no checked habitat for 14 of the 15 menu dinosaurs. Spinosaurus is river floodplain (rivers and deltas in what is now the Sahara).
- **Coelophysis dates.** A 2024 study puts it at ~206–201 (Rhaetian). Older books give ~225. Accept either.
- **Eoraptor.** Sources disagree on diet (Sereno lab: meat eater) and on whether it is an early theropod or an early sauropodomorph. It is lizard-hipped either way. Accept carnivore or omnivore.
- **Liliensternus** is the weakest menu slot: its dates are poorly pinned. Accept any Late Triassic date.
- **Walks on.** 4 legs for Stegosaurus, the sauropods, Triceratops and Ankylosaurus is standard textbook knowledge. It was not separately re-sourced.
- **Map dots not verified.** T. rex in North Dakota and Alberta, and Triceratops in Colorado, are likely but were not confirmed. A dot there is **PROVISIONAL**. Decide before grading whether it costs points.
- **Name meanings** are left off the menu for Herrerasaurus, Coelophysis, Allosaurus and Liliensternus. They were not verified. Name meaning is not graded.
- **No paleomap is supplied.** Students draw or trace their own. The map row grades “not today's map” and dots in the right places, not accurate coastlines. Open question for a class reference map (report §4): Blakey's 90 Ma Commons map (CC BY-SA 3.0, low-res only) or a trace from Scotese & Wright 2018 PALEOMAP.
- **School bus.** A full-size Type C bus is 12.0 m (39.3 ft). Re-solved: T. rex 12–13 m is about 1 bus; Brachiosaurus 18–22 m is 1.5 to 1.8 buses; Diplodocus 24–26 m is 2 to 2.2 buses; Velociraptor 1.5–2 m is an eighth to a sixth of a bus.

SPINOSAURUS NOTES

- **Swim or wade? Unresolved.** Everyone agrees it lived around water and ate fish. Ibrahim et al. 2014 (Science) and 2020 (Nature, the fin-like tail) and Fabbri et al. 2022 (dense, diver-like bones) argue it hunted underwater. Henderson 2018, Hone & Holtz 2021 (a wading “shoreline generalist”, like a giant heron), Sereno et al. 2022 (“not an aquatic dinosaur”) and Myhrvold et al. 2024 (Fabbri's statistics unreliable; Fabbri's team says its result holds) argue it waded and ambushed from the shore. Student-safe line: “Scientists are still arguing about whether it swam underwater to hunt or waded in the shallows like a giant heron.”
- **Holotype destroyed.** BSP 1912 VIII 19, found in Egypt's Bahariya Formation in 1912 and named by Stromer in 1915, was destroyed on the night of 24/25 April 1944 when a British bombing raid hit the Munich museum. Only Stromer's descriptions, drawings and a photograph survive. The Moroccan neotype (FSAC-KK 11888, Ibrahim et al. 2014) has a disputed status.
- **S. mirabilis, 2026.** A second species named in February 2026 from Niger's Farak Formation (~95 Ma; Sereno et al., Science 391). **PROVISIONAL**: one paper, not yet independently tested. The menu treats Spinosaurus as a genus, so a Niger dot is acceptable. A student who leaves Niger off because they mean S. aegyptiacus is also right.
- **Age.** The rocks are Cenomanian, the first stage of the Late Cretaceous. “Middle” on the menu is the informal “mid-Cretaceous” label papers also use. Accept Middle or Late Cretaceous.
- **Posture.** The menu says 2 legs. The 4-leg reconstruction (Ibrahim 2014) is older and now mostly rejected.
- **Size.** Length estimates run 12.6 to 18 m; the menu uses 12 to 15. Mass estimates run from about 6 to 21 t, because no complete skeleton exists. Say “one of the longest meat-eating dinosaurs known.” Do not let “the biggest meat-eater ever” onto a poster as fact.

COMMON MISTAKES

STUDENTS SAY OR DRAW	CORRECTION
“Pterosaurs are flying dinosaurs.”	Not dinosaurs. They are the dinosaurs' closest relatives.
“Plesiosaurs, ichthyosaurs and mosasaurs are swimming dinosaurs.”	Marine reptiles, not dinosaurs. Mosasaurs are lizards (squamates).
“Dimetrodon is a dinosaur.”	A Permian synapsid, closer to mammals. It died out roughly 60 million years before the first dinosaurs.
Grassy plains behind the dinosaur.	Grass was absent or rare for most of the Mesozoic. Grasslands spread in the Cenozoic. Use ferns, conifers, cycads, and flowering plants in the Cretaceous.
T. rex fighting Stegosaurus.	At least 77 million years apart (~145–150 vs ~68 mya). T. rex lived closer in time to us than to Stegosaurus.
Spinosaurus fighting T. rex.	At least ~26 million years apart (93.9 – 68), and on different continents.

Movie-sized Velociraptor with scales.	Turkey-sized (~1.5–2 m, mostly tail) and feathered: quill knobs on its forearm (Turner et al. 2007). Movie raptors are closer to Deinonychus.
Dilophosaurus with a frill that spits venom.	No evidence for either. The real animal was bigger than the movie one, ~6–7 m.
“Bird-hipped dinosaurs became birds.”	No. Birds came from lizard-hipped theropods.
Mussaurus as a Triassic dinosaur.	Now dated to the Early Jurassic, ~193 mya. Older books say Triassic.
The Berlin “Brachiosaurus” skeleton (Tanzania).	Giraffatitan since Taylor 2009. Real Brachiosaurus is North American. No Africa dot.
Dinosaurs and people at the same time.	Non-bird dinosaurs died out 66 mya. Homo sapiens is about 300,000 years old.
Hip type judged from a Velociraptor or bird skeleton.	Pubis points backward even though it is lizard-hipped.
One “official” length or weight for Spinosaurus.	Estimates run 12–18 m and 6–21 t. Teach the range.
“All dinosaurs are extinct.”	Birds are living dinosaurs. All dinosaurs except birds died out 66 mya.